

Which Reduced Hypotenuses Arise from Three Pythagorean Points?

An Elliptic Envelope, a Global Certificate Height, and Two Realizability Questions

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Abstract

Which primitive Pythagorean hypotenuses can arise when three primitive integral points on the Pythagorean cone are joined by integer distances and the resulting right triangle is reduced to primitive form? The search begins with two small clues, 17 and 41, followed by a striking gap. A direct parameter cutoff cannot certify that gap, because very large spatial configurations may collapse to small primitive distance triangles after division by a large common factor. The problem therefore forces a change of viewpoint from size to shape.

Following that trail reveals three related arithmetic-geometric layers. A special linear two-parameter construction gives OEIS A399309 and leads to a genus-one quartic and a rank-3 elliptic curve. Keeping every primitive Pythagorean hypotenuse supplied by rational points on that quartic gives the larger arithmetic sequence A399677. Abandoning the special construction and allowing arbitrary triples of primitive Pythagorean points gives the global geometric sequence A397123, ordered by a scale-free minimal certificate height for which every bounded prefix is finite and exhaustive.

Two exact separations show that these layers do not collapse: 521 belongs to the quartic envelope but not to the original special family, while 493 is globally realizable yet is excluded from the quartic by an elementary local prime-divisor obstruction. Two natural realizability questions remain. First, is every quartic hypotenuse globally realizable? The first unresolved quartic target is 901; if the conjectured inclusion holds, its minimal global certificate height exceeds 20000. Second, does the global value set equal the classical set of all primitive Pythagorean hypotenuses, OEIS A008846? This broader equality is not known. Thus A397123 should be viewed unconditionally as a geometric complexity ordering; whether it also selects a proper subset of A008846 remains open.

This paper develops the exact algebra behind these layers, isolates the mod-8 shadow recorded by A399705, and separates proved statements from the two open universality questions. Exhaustive statements involving the elliptic curve use finite exact computer-assisted certificates (Magma and Maple), while the global certificate-completeness theorem is proved algebraically and implemented in exact Maple arithmetic. No analytic-rank or BSD assumption is used.

1 The search begins: which hypotenuses can occur?

The guiding question is deliberately elementary to state:

which primitive Pythagorean hypotenuses can occur as reduced distance hypotenuses?

At first sight this looks like an enumeration problem. It is not. The first examples are small, but a large spatial configuration can have a very large common distance divisor and therefore

reduce to a much smaller primitive triangle. A cutoff in the original Euclid parameters can find examples, but it cannot prove that an unseen smaller reduced hypotenuse does not occur. The search must therefore forget absolute scale and retain only rational shape.

Every primitive Pythagorean triple is generated by a canonical Euclid pair (m, k) with

$$m > k > 0, \quad \gcd(m, k) = 1, \quad m - k \text{ odd.}$$

Instead of using the three Euclid expressions as side lengths, we use them as coordinates:

$$P(m, k) = (m^2 - k^2, 2mk, m^2 + k^2) \in \mathbb{Z}^3. \quad (1)$$

Thus $P(m, k)$ is an integral point on $x^2 + y^2 = z^2$.

Definition 1.1. Three distinct primitive canonical Pythagorean points are called *admissible* if their three pairwise Euclidean distances are nonzero integers and form a nondegenerate right triangle. Dividing the three distances by their common gcd gives a primitive Pythagorean triangle. Its hypotenuse is the *reduced hypotenuse* of the realization.

The investigation proceeds by widening the search in three stages.

Layer 1. The special linear family. The three Euclid pairs are fixed linear forms in two odd coprime parameters u, v . The corresponding reduced hypotenuses form A399309.

Layer 2. The quartic arithmetic envelope. Forget the original linear parameters but retain the genus-one quartic forced by the leg-ratio calculation. Every rational quartic point gives a primitive Pythagorean hypotenuse. These form A399677.

Layer 3. The global geometry. Allow arbitrary triples from Definition 1.1. Their reduced hypotenuses form the value set underlying A397123. The sequence itself is ordered by minimal scale-free certificate height rather than by numerical size.

Let $\mathcal{S}_{309}$, $\mathcal{S}_{677}$ and $\mathcal{S}_{123}$ denote the sets of values occurring in A399309, A399677 and A397123, respectively. Let

$$\mathcal{H} = \{\text{values occurring in A008846}\}$$

be the set of all primitive Pythagorean hypotenuses. By definition,

$$\mathcal{S}_{123} \subseteq \mathcal{H}. \quad (2)$$

The distinction between a value set and a sequence order matters here: A397123 is ordered by minimal certificate height rather than numerically. We shall prove

$$\mathcal{S}_{309} \subsetneq \mathcal{S}_{677}, \quad \mathcal{S}_{309} \subsetneq \mathcal{S}_{123}, \quad \mathcal{S}_{123} \not\subseteq \mathcal{S}_{677}.$$

Within these three layers, the remaining inclusion

$$\mathcal{S}_{677} \stackrel{?}{\subseteq} \mathcal{S}_{123}$$

is open. A second, broader question is whether (2) is strict or instead $\mathcal{S}_{123} = \mathcal{H}$. The latter question does not affect the definition of A397123: even in the equality case, A397123 records the new geometric complexity invariant $h(C)$ through its ordering.

2 First clues: 17, 41, and the special family A399309

2.1 The first mechanism: a fixed orthogonal construction

Let u, v be coprime positive odd integers and put

$$(m_0, k_0) = \left(\frac{17v - 7u}{2}, \frac{21u - 17v}{2} \right), \quad (3)$$

$$(m_1, k_1) = \left(\frac{85v - 77u}{2}, \frac{51v - 49u}{2} \right), \quad (4)$$

$$(m_2, k_2) = \left(\frac{17u - 7v}{2}, \frac{21v - 17u}{2} \right). \quad (5)$$

Retain only those u, v for which all three pairs are canonical Euclid pairs, and set $P_i = P(m_i, k_i)$.

Direct expansion gives

$$P_1 - P_0 = 2(17v - 14u)(34v - 35u)(1, 2, 2), \quad (6)$$

$$P_2 - P_0 = (u^2 - v^2)(98, -71, 22). \quad (7)$$

The direction vectors are orthogonal because

$$(1, 2, 2) \cdot (98, -71, 22) = 98 - 142 + 44 = 0.$$

Hence every retained parameter pair produces a right triangle at P_0 . Since $\|(1, 2, 2)\| = 3$ and $98^2 + 71^2 + 22^2 = 123^2$, the two leg lengths are

$$D_1 = 6|(17v - 14u)(34v - 35u)|, \quad D_2 = 123|u^2 - v^2|. \quad (8)$$

The first two examples reduce to $(8, 15, 17)$ and $(9, 40, 41)$.

2.2 The hidden scale-free quartic

Put $t = v/u$. From (8), up to sign and leg interchange,

$$\frac{D_1}{D_2} = \frac{2(17t - 14)(34t - 35)}{41(t^2 - 1)}. \quad (9)$$

Let r, w be coprime Euclid parameters for the reduced distance triangle and put $s = r/w$. A signed leg ratio is

$$\frac{2s}{1 - s^2}. \quad (10)$$

Equating the two ratios gives

$$(578w^2 - 41rw - 578r^2)t^2 + 1071(r^2 - w^2)t + (490w^2 + 41rw - 490r^2) = 0. \quad (11)$$

Rationality of t forces the discriminant to be a square, giving

$$\boxed{z^2 = 14161r^4 + 14432r^3w - 21598r^2w^2 - 14432rw^3 + 14161w^4}. \quad (12)$$

Conversely, a rational point of (12) gives at most two rational candidates for t through (11); the original positivity, coprimality and parity conditions decide exactly whether either candidate lies in the special family.

For an admissible special-family configuration, u, v are odd. Multiplying (11) by u^2 and reducing modulo 2 gives $r^2 - w^2 \equiv 0 \pmod{2}$. Since $\gcd(r, w) = 1$, both r, w are odd and

$$C = \frac{r^2 + w^2}{2}. \quad (13)$$

2.3 The hidden elliptic curve

Set $s = r/w$ and $z_0 = z/w^2$. Then (12) becomes

$$z_0^2 = 14161s^4 + 14432s^3 - 21598s^2 - 14432s + 14161. \quad (14)$$

The rational point $(0, 119)$ makes this genus-one quartic an elliptic curve after choosing an origin. An explicit Weierstrass model is

$$E : y^2 = x^3 - x^2 - 72869344x + 237569291584 = (x + 9848)(x - 4568)(x - 5281). \quad (15)$$

For $s \neq 0$, one forward map is

$$x = \frac{-10797s^2 - 21648s + 357z_0 + 42483}{6s^2}, \quad (16)$$

$$y = \frac{429352s^3 - 1285081s^2 - 3608sz_0 - 1288056s + 14161z_0 + 1685159}{2s^3}. \quad (17)$$

For the inverse map put $x' = 9x - 3$, $y' = 27y$. Away from a finite exceptional set,

$$s = \frac{3(14161x' - 575932893)}{10824x' + 119y' - 514421424}, \quad (18)$$

$$z_0 = -119 + \frac{s(2s(2x' + 32397)/9 + 14432)}{238}. \quad (19)$$

Direct substitution verifies both directions.

Lemma 2.1 (Exceptional fibers). *No admissible special-family configuration is lost by the birational formulas (16)–(19).*

Proof. The forward formulas have $s = 0$ as their only affine pole; this means $r = 0$ and is excluded for Euclid parameters. The inverse denominator vanishes on

$$10824x' + 119y' - 514421424 = 0.$$

Intersecting this line with the intermediate cubic gives

$$(x' - 47526)(x' + 79923)(14161x' - 575932893) = 0.$$

The first two intersections are the two points at infinity of the projective quartic, hence $w = 0$, while the third is the exceptional affine point above $s = 0$. The remaining point at infinity accounts for the other point above $s = 0$. None represents a reduced Pythagorean triangle. $\square$

2.4 Certifying that the elliptic search is complete

Three quartic points arising from the first three special-family hypotenuses map to

$$\begin{aligned} G_1 &= (20410, 2692962), \\ G_2 &= \left(-\frac{770792}{81}, \frac{191768480}{729} \right), \\ G_3 &= (13768, 1357920). \end{aligned}$$

The exact torsion computation gives

$$E(\mathbb{Q})_{\text{tors}} = \{O, (-9848, 0), (4568, 0), (5281, 0)\} \cong (\mathbb{Z}/2\mathbb{Z})^2. \quad (20)$$

The canonical-height pairing matrix of G_1, G_2, G_3 is approximately

$$\begin{pmatrix} 4.4000420535 & 3.3504649357 & -2.1694454235 \\ 3.3504649357 & 5.5951965504 & -0.7414509197 \\ -2.1694454235 & -0.7414509197 & 2.6468581162 \end{pmatrix}.$$

The exact Maple interval certificate proves

$$\widehat{h}(j_1G_1 + j_2G_2 + j_3G_3 + T) > \frac{16}{25}(j_1^2 + j_2^2 + j_3^2) \quad (21)$$

for every non-torsion point in this subgroup, and also

$$\text{Reg}\langle G_1, G_2, G_3 \rangle < 18. \quad (22)$$

Hence G_1, G_2, G_3 are independent.

An algebraic 2-isogeny/Selmer descent in Magma gives

$$\text{RankBounds}(E) = [3, 3], \quad (23)$$

so $\text{rank } E(\mathbb{Q}) = 3$. No analytic-rank assumption is used.

To prove that the full-rank subgroup has index 1, we need a lower bound for all non-torsion points of $E(\mathbb{Q})$. Let

$$x(Q) = A/b^2$$

be in lowest terms and put $h(Q) = \log \max(|A|, b^2)$. The Cremona–Prickett–Siksek height-difference bound, used with the safe rational constant 13, implies that after adding a suitable torsion point,

$$h(Q + T) - \widehat{h}(Q) < 13. \quad (24)$$

Proposition 2.2 (Certified small-height audit). *Every non-torsion $Q \in E(\mathbb{Q})$ satisfies*

$$\widehat{h}(Q) > \frac{3}{250}.$$

Computer-assisted exact proof. If $\widehat{h}(Q) \leq 3/250$, then by (24), after a torsion translation,

$$h(Q + T) < 13.012.$$

Thus, writing $x(Q + T) = A/b^2$ in lowest terms,

$$|A| \leq 524287, \quad 1 \leq b \leq 724. \quad (25)$$

For the factorized model (15), such an x -coordinate lies on E iff

$$(A + 9848b^2)(A - 4568b^2)(A - 5281b^2) \quad (26)$$

is a nonnegative integer square. The Maple certificate exhausts every pair (A, b) in (25); congruence filters only discard impossible squares, and every survivor is checked by an exact integer square test. It finds exactly 32 rational x -values: three finite 2-torsion values and 29 non-torsion values. Repeated doubling with the same safe height-difference bound certifies each of the 29 values above $3/250$. The recorded decisive output is

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curve x-values = 32,
non-torsion x-values certified = 29,
torsion x-values = 3,
unresolved x-values = 0.
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Thus no non-torsion point with canonical height at most $3/250$ exists. $\square$

Let $L = \langle G_1, G_2, G_3 \rangle$ and $I = [E(\mathbb{Q})_{\text{free}} : L]$. Exact Magma saturation proves that no prime $p \leq 5000$ divides I ; the 2-saturation is checked separately and all odd primes up to 5000 are then saturated. On the other hand, the Hermite constant in dimension 3 is $\gamma_3 = 2^{1/3} < 63/50$. Proposition 2.2 and (22) give

$$\begin{aligned} I^2 &= \frac{\text{Reg}(L)}{\text{Reg}(E(\mathbb{Q})_{\text{free}})} \\ &< 18 \left(\frac{63/50}{3/250} \right)^3 = 20837250 < 5000^2. \end{aligned}$$

Hence $I < 5000$. If $I > 1$, it has a prime divisor below 5000, contradicting saturation. Therefore $I = 1$.

Theorem 2.3 (Computer-assisted Mordell–Weil theorem).

$$E(\mathbb{Q}) = \langle G_1, G_2, G_3 \rangle + E(\mathbb{Q})_{\text{tors}}, \quad E(\mathbb{Q})_{\text{tors}} \cong (\mathbb{Z}/2\mathbb{Z})^2.$$

Every rational point is therefore

$$j_1 G_1 + j_2 G_2 + j_3 G_3 + T, \quad j_i \in \mathbb{Z}, \quad T \in E(\mathbb{Q})_{\text{tors}}.$$

Remark 2.4 (Reproducibility of the computer-assisted input). The computer-assisted parts of Theorems 2.3 and 2.5 are supported by the exact certificate archive accompanying this paper [1], reserved at Zenodo DOI 10.5281/zenodo.22666219. The archive contains the Magma rank and saturation scripts, the Maple height and finite-audit certificates, the global A397123 implementation, and a SHA-256 manifest. The proof uses finite algebraic and rational/integer computations, not numerical evidence.

2.5 Turning the infinite search into a finite box

Let

$$H(r, w) = \frac{r^2 + w^2}{\gcd(w^2 - r^2, 2rw, r^2 + w^2)}.$$

For coprime r, w , the denominator is 1 in the opposite-parity case and 2 in the odd–odd case. Hence with $S = r^2 + w^2$,

$$S \leq 2H(r, w). \tag{27}$$

Homogenizing (16) gives

$$x(P) = \frac{-10797r^2 - 21648rw + 357z + 42483w^2}{6r^2}.$$

The quartic gives $|z|^2 \leq 78784S^2 < 281^2 S^2$, so the numerator has absolute value at most $164421S$. Therefore, if $H(r, w) \leq N$,

$$h(P) \leq \log(328842N). \tag{28}$$

A direct duplication-formula estimate on (15) gives

$$\hat{h}(P) \leq h(P) + \frac{\log M}{3}, \quad M = 5312792272228033 < 10^{16}. \tag{29}$$

If N has d decimal digits, then $328842 < 10^6$ and $\log 10 < 2303/1000$ yield

$$\hat{h}(P) < \frac{2303(3d + 34)}{3000}. \tag{30}$$

Theorem 2.5 (Finite Mordell–Weil box with certified computer-assisted input). *Let N have d decimal digits and let L be the least positive integer with*

$$1920L^2 \geq 2303(3d + 34). \quad (31)$$

Every quartic value $H(r, w) \leq N$ arises from a point

$$j_1G_1 + j_2G_2 + j_3G_3 + T$$

with

$$-L + 1 \leq j_i \leq L - 1 \quad (i = 1, 2, 3)$$

and T torsion. Enumerating this finite box, applying (18)–(19), and testing the desired arithmetic or geometric conditions exactly is exhaustive.

Proof. If some $|j_i| \geq L$, then (21) gives $\hat{h}(P) > (16/25)L^2$, which is at least the upper bound (30) by (31). This is impossible. The inverse map and subsequent tests use exact rational/integer arithmetic, so no retained point is spurious and no point satisfying the requested filter is omitted. $\square$

For A399309 the final filter is exactly the original reconstruction of $t = v/u$ and the three canonical Euclid conditions. This proves, in particular, that its beginning is

$$17, 41, 5867801, 7362241, 24793297, \dots,$$

so there is no special-family value between 41 and 5867801.

2.6 A first rigid fingerprint: the mod-8 law

The special family has an additional local rigidity not shared by the whole quartic. We give the full argument because it is used later to distinguish the layers.

Write

$$T_1 = (-9848, 0), \quad T_2 = (4568, 0), \quad T_3 = (5281, 0), \quad L = \langle G_1, G_2, G_3 \rangle.$$

Lemma 2.6 (Admissible torsion cosets). *Every admissible special-family point lies in $L + \{O, T_3\}$.*

Proof. Modulo 17, the factorized model is

$$y^2 = (x - 12)^2(x - 11),$$

with singular point $(12, 0)$. The reductions

$$G_1 \mapsto (10, 9), \quad G_2 \mapsto (3, 10), \quad G_3 \mapsto (15, 11), \quad T_3 \mapsto (11, 0)$$

are nonsingular, whereas T_1, T_2 reduce to $(12, 0)$. Thus $L + \{O, T_3\}$ lies in the nonsingular component and the other two torsion cosets lie in the singular component.

For a point reducing to $(12, 0)$, formula (18) gives $s \equiv 13 \pmod{17}$. In affine form (11) is

$$(-578s^2 - 41s + 578)t^2 + 1071(s^2 - 1)t + (-490s^2 + 41s + 490) = 0.$$

At $s \equiv 13 \pmod{17}$ the three coefficients are 11, 0, 0 modulo 17. The leading coefficient is a 17-adic unit, so any rational root is 17-integral; reduction gives $t \equiv 0 \pmod{17}$. Writing $t = v/u$ in lowest terms gives $17 \mid v$ and $17 \nmid u$. Then the third Euclid pair

$$\left(\frac{17u - 7v}{2}, \frac{21v - 17u}{2} \right)$$

has both entries divisible by 17, contradicting primitivity. $\square$

Let $\chi(a) = (2, a)_2$ be the 2-adic Hilbert-symbol character. If $a = 2^n q$ with q an odd 2-adic unit, then

$$\chi(a) = +1 \text{ for } q \equiv 1, 7 \pmod{8}, \quad \chi(a) = -1 \text{ for } q \equiv 3, 5 \pmod{8}.$$

The Kummer map associated with T_1 is represented away from T_1 by $x(P) + 9848$ in $\mathbb{Q}^*/\mathbb{Q}^{*2}$, with the standard special definitions at O, T_1 . Hence

$$\varepsilon(P) = \chi(x(P) + 9848)$$

is a homomorphism. On the generators of the admissible subgroup,

$$\begin{aligned} x(G_1) + 9848 &= 30258 = 2 \cdot 123^2, \\ x(G_2) + 9848 &= 26896/81 = (164/9)^2, \\ x(G_3) + 9848 &= 23616 = 24^2 \cdot 41, \\ x(T_3) + 9848 &= 15129 = 123^2. \end{aligned}$$

Since $41 \equiv 1 \pmod{8}$, all four have character $+1$. Thus

$$\varepsilon(P) = +1 \tag{32}$$

for every admissible point.

Now let P correspond to (s, z_0) and $\bar{P}$ to $(s, -z_0)$. Put $q = x(P) + 9848$ and $\bar{q} = x(\bar{P}) + 9848$. The forward map and the quartic give the exact identities

$$q + \bar{q} = \frac{16097s^2 - 7216s + 14161}{s^2}, \tag{33}$$

$$q\bar{q} = 8 \left(\frac{123(11s - 41)}{s} \right)^2. \tag{34}$$

For an admissible point, s is a 2-adic unit. The numerator in (33) is congruent to $s^2 + 1 \pmod{16}$ and therefore has valuation exactly 1. The exceptional fiber $s = 41/11$ contains a 2-torsion point and its two reconstructed t -values fail the special-family Euclid conditions, so it is not admissible. Hence (34) has odd valuation at least 5. The valuations of $q, \bar{q}$ are therefore unequal; one is 1 and the other at least 4.

Because (34) is 8 times a square, $\chi(q)\chi(\bar{q}) = \chi(8) = 1$, so $q, \bar{q}$ have the same character. The higher-valuation term changes the odd unit part of the lower-valuation term in the sum only by a multiple of 8, so this common character is the character of $q + \bar{q}$. Therefore

$$\varepsilon(P) = \begin{cases} +1, & s \equiv \pm 1 \pmod{8}, \\ -1, & s \equiv \pm 3 \pmod{8}. \end{cases}$$

Combining this with (32) gives $s \equiv \pm 1 \pmod{8}$.

Theorem 2.7 (Special-family congruence). *Every $C \in \mathcal{S}_{309}$ satisfies*

$$\boxed{C \equiv 1 \pmod{8}}.$$

More precisely, if $s = r/w$ is in lowest terms, then $r \equiv \pm w \pmod{8}$.

Proof. The preceding character calculation gives $s \equiv \pm 1 \pmod{8}$, hence $r \equiv \pm w \pmod{8}$ because r, w are odd. Thus $r^2 \equiv w^2 \pmod{16}$, and (13) gives $C \equiv 1 \pmod{8}$. $\square$

3 Widening the net: the arithmetic quartic envelope A399677

Once the special family is understood, the next question is whether its linear construction is essential or merely one route onto the quartic. We therefore remove the original reconstruction filter but keep the quartic itself. This is the first widening of the search.

Definition 3.1. Let r, w be coprime nonzero integers with $r^2 \neq w^2$, and suppose that (12) holds for an integer z . Define

$$H(r, w) = \frac{r^2 + w^2}{\gcd(w^2 - r^2, 2rw, r^2 + w^2)}.$$

The set $\mathcal{S}_{677}$ consists of all distinct positive values $H(r, w)$.

Since $\gcd(r, w) = 1$,

$$H(r, w) = \begin{cases} r^2 + w^2, & r, w \text{ of opposite parity,} \\ (r^2 + w^2)/2, & r, w \text{ both odd.} \end{cases} \quad (35)$$

Thus every element is a primitive Pythagorean hypotenuse. Every special-family output determines a quartic point, so

$$\mathcal{S}_{309} \subseteq \mathcal{S}_{677}.$$

The inclusion is strict: the exact quartic point

$$(r, w, z) = (-20, 11, 23279)$$

gives

$$H = 20^2 + 11^2 = 521,$$

whereas the exhaustive special-family theorem above shows $521 \notin \mathcal{S}_{309}$.

Theorem 2.5 applies without the special-family admissibility filter and therefore yields a provably exhaustive numerical enumeration of A399677 below every prescribed bound. Its first terms are

$$17, 41, 521, 881, 901, 3457, 7577, 15317, 29857, 36941, \dots$$

For example, 901 comes from the exact quartic point

$$(r, w, z) = (41, 11, 214438), \quad 901 = \frac{41^2 + 11^2}{2}.$$

3.1 A local fingerprint: the exceptional primes 17 and 53

Proposition 3.2 (Prime divisors of quartic hypotenuses). *Let $H \in \mathcal{S}_{677}$ and let p be an odd prime divisor of H . If $p \notin \{17, 53\}$, then*

$$p \equiv 1 \pmod{8}.$$

Proof. Every prime divisor of a primitive Pythagorean hypotenuse is $1 \pmod{4}$. Write $H = H(r, w)$. Since $p \mid H$, one has $r^2 + w^2 \equiv 0 \pmod{p}$. Coprimality implies $p \nmid w$, so put $i = r/w \pmod{p}$. Then $i^2 = -1$. Dividing the quartic by w^4 and using $i^2 = -1$ gives, up to $i \mapsto -i$,

$$(z/w^2)^2 \equiv 49920 - 28864i = 32(41 - 11i)^2 \pmod{p}.$$

If $41 - 11i \equiv 0 \pmod{p}$, then $i^2 = -1$ implies

$$p \mid 41^2 + 11^2 = 1802 = 2 \cdot 17 \cdot 53.$$

For $p \notin \{17, 53\}$ the factor is nonzero, so 32 is a square modulo p . Since $32 = 2^5$, this means $(2/p) = 1$. Together with $p \equiv 1 \pmod{4}$, the supplementary law for 2 forces $p \equiv 1 \pmod{8}$. $\square$

Corollary 3.3 (The mod-8 bit inside the quartic layer). *For $H \in \mathcal{S}_{677}$,*

$$H \equiv 5 \pmod{8} \iff \nu_{53}(H) \text{ is odd.}$$

Proof. All prime divisors are 1 or 5 (mod 8). Proposition 3.2 shows that the only possible 5 (mod 8) prime is 53; $17 \equiv 1 \pmod{8}$. Multiplying residue classes gives the claim. $\square$

4 Leaving the quartic: the global search A397123

The quartic is a natural candidate for the hidden arithmetic core of the problem, but it might still be only one component of a larger geometry. To test that possibility we now abandon the quartic as a defining restriction and search among all admissible triples. The price is a much larger parameter space; the reward is a certificate height that makes every bounded part of that space finite and exact.

4.1 Normal form for one edge

For Euclid vectors $a = (m, k)$ and $b = (m', k')$, direct expansion gives

$$(P(a) - P(b))_1^2 + (P(a) - P(b))_2^2 - (P(a) - P(b))_3^2 = 4 \det(a, b)^2. \quad (36)$$

Thus an admissible edge direction has both Euclidean and Minkowski norms equal to rational squares.

For a primitive projective pair $[p : q]$ with $p > 0$, $q \geq 0$ and $\gcd(p, q) = 1$, define

$$d(p, q) = (p^2, 2q^2, 2pq). \quad (37)$$

Its two quadratic norms are

$$X^2 + Y^2 + Z^2 = (p^2 + 2q^2)^2, \quad X^2 + Y^2 - Z^2 = (p^2 - 2q^2)^2.$$

Lemma 4.1 (Normal form for an admissible direction). *Up to rational scaling and a rational rotation about the z -axis, every rational direction whose Euclidean and Minkowski norms are rational squares has the form (37).*

Proof. Scale so that the Euclidean norm is 1. If z is the third coordinate and k the Minkowski norm, then $k^2 + 2z^2 = 1$. The rational points of this conic are

$$z = \frac{2pq}{p^2 + 2q^2}, \quad k = \frac{p^2 - 2q^2}{p^2 + 2q^2}.$$

The normalized vector $d(p, q)/(p^2 + 2q^2)$ has the same third coordinate and same squared horizontal norm. The two horizontal vectors therefore differ by a rational planar rotation: its cosine and sine are their rational dot product and determinant divided by the common rational squared horizontal norm. $\square$

4.2 Making two directions orthogonal

Take $d(p, q)$ fixed and rotate $d(u, v)$ about the z -axis by θ . Put

$$A_0 = p^2 u^2 + 4q^2 v^2, \quad B_0 = 2(q^2 u^2 - p^2 v^2), \quad C_0 = 4pquv,$$

and $R = A_0^2 + B_0^2$. Orthogonality is

$$A_0 \cos \theta + B_0 \sin \theta + C_0 = 0.$$

Intersecting this line with $\cos^2 \theta + \sin^2 \theta = 1$ gives rational solutions exactly when

$$W^2 = R - C_0^2 = (p^2 u^2 - 4q^2 v^2)^2 + 4(p^2 v^2 - q^2 u^2)^2 \quad (38)$$

is a square. The two rational branches are

$$\cos \theta_\varepsilon = \frac{-A_0 C_0 - \varepsilon B_0 W}{R}, \quad (39)$$

$$\sin \theta_\varepsilon = \frac{-B_0 C_0 + \varepsilon A_0 W}{R}, \quad (40)$$

for $\varepsilon = \pm 1$. Thus (38) is an exact necessary-and-sufficient orthogonality test for a fixed direction pair.

A primitive target pair (r, s) with

$$r > s > 0, \quad \gcd(r, s) = 1, \quad r - s \text{ odd}$$

represents the primitive triangle

$$(A, B, C) = (r^2 - s^2, 2rs, r^2 + s^2) \quad (41)$$

up to leg interchange.

There is also a useful exact rejection test before reconstruction. Put

$$L_1 = p^2 + 2q^2, \quad z_1 = 2pq, \quad L_2 = u^2 + 2v^2, \quad z_2 = 2uv, \quad D = L_1 L_2.$$

For one of the assignments $(a, b) = (A, B)$ or (B, A) and a sign $\eta = \pm 1$, set

$$T = az_1 L_2 + \eta bz_2 L_1. \quad (42)$$

Rational target compatibility requires

$$N = C^2 D^2 - 2T^2 \quad (43)$$

to be a nonnegative square. This is the denominator-cleared identity $Y^2 = C^2 - 2(a\alpha - b\beta)^2$ with $\alpha = z_1/L_1$, $\beta = z_2/L_2$. It is used only as a necessary rejection test.

4.3 Reconstructing a candidate exactly

Let $d = (X, Y, Z)$ be a rational direction with

$$L^2 = X^2 + Y^2 + Z^2, \quad K^2 = X^2 + Y^2 - Z^2,$$

where $L, K > 0$. Define

$$F_d(m, k) = (X - Z)m^2 + 2Ymk - (X + Z)k^2 \quad (44)$$

and

$$M_d = \frac{1}{K} \begin{pmatrix} -Y & X + Z \\ X - Z & Y \end{pmatrix}. \quad (45)$$

Lemma 4.2 (Edge involution and exact length). *One has $M_d^2 = I$. If $(m', k') = M_d(m, k)$, then*

$$P(m', k') - P(m, k) = -\frac{2F_d(m, k)}{K^2} d. \quad (46)$$

Consequently the Euclidean edge length is

$$D_d(m, k) = \frac{2L}{K^2} |F_d(m, k)|. \quad (47)$$

Proof. Multiplying the matrix in (45) by itself and using $K^2 = X^2 + Y^2 - Z^2$ gives $M_d^2 = I$. Substituting

$$m' = \frac{-Ym + (X + Z)k}{K}, \quad k' = \frac{(X - Z)m + Yk}{K}$$

into (1) and simplifying gives (46). Taking Euclidean norms gives (47). $\square$

For two rational orthogonal directions $d_i = (X_i, Y_i, Z_i)$, with lengths L_i, K_i , a target-leg assignment (A, B) and a sign $\sigma = \pm 1$, the required edge-length ratio is equivalent to the homogeneous quadratic equation

$$BL_1K_2^2F_{d_1}(m, k) - \sigma AL_2K_1^2F_{d_2}(m, k) = 0. \quad (48)$$

All rational roots are enumerated exactly. From a root (m, k) , the three rational spinors are

$$(m, k), \quad M_{d_1}(m, k), \quad M_{d_2}(m, k),$$

up to the finitely many sign and branch choices already enumerated. Lemma 4.2 then gives the two prescribed orthogonal edge directions and the target leg ratio.

The last two filters are also exact, rather than numerical angle or gcd tests.

Lemma 4.3 (Gaussian normalization). *Let three rational spinors be fixed by one reconstruction branch. Clear a common rational denominator, obtaining $z_0, z_1, z_2 \in \mathbb{Z}[i]$, and let g be a Gaussian gcd of the three. A common Gaussian rational scaling can make all three spinors integral primitive Euclid pairs if and only if, after division by g , each spinor has coprime real and imaginary parts of opposite parity.*

Proof. After dividing by g , the three Gaussian integers have no common Gaussian prime divisor. Hence any further common Gaussian rational multiplier that keeps all three integral has Gaussian-integer denominator a unit; it is therefore a Gaussian integer up to a unit. If one normalized spinor has a common rational prime divisor in its two coordinates, or has equal parity (equivalently is divisible by $1 + i$), multiplication by a Gaussian integer cannot remove that divisibility. Conversely, if all three normalized spinors already have coprime coordinates of opposite parity, the identity multiplier gives three primitive Euclid spinors. Thus the test is necessary and sufficient. $\square$

Lemma 4.4 (Exact canonical chamber test). *For three nonzero integer spinors $z_i = (m_i, k_i)$, independent sign changes and a common planar rotation place all three in the open chamber $0 < k < m$ if and only if, for all pairs $i < j$,*

$$|z_i \cdot z_j| > |\det(z_i, z_j)|$$

and the product of the three pairwise dot products is positive.

Proof. For two nonzero planar vectors with angle difference δ , $|\tan \delta| = |\det|/|\text{dot}|$. Thus the strict inequality says that the corresponding unoriented lines differ by less than $\pi/4$. The positive product of the three dot products is exactly the consistency condition that independent signs can orient the three vectors so that all pairwise dot products are positive. Their three oriented angles then lie in one open interval of width $\pi/4$, and a common rotation moves that interval into $(0, \pi/4)$. The converse is immediate. $\square$

The implementation therefore clears denominators, divides by the common Gaussian gcd, applies the primitive-parity test of Lemma 4.3, and applies the dot-product/determinant test of Lemma 4.4; all operations are exact integers. Finally the three Pythagorean points are formed and their distances are checked directly against Definition 1.1. Thus every accepted output is independently verified at the original geometric level.

4.4 A height that makes the global search finite

Definition 4.5. A global certificate consists of two primitive projective direction parameters $[p : q], [u : v]$, a primitive target pair (r, s) , and one of the finitely many rotation, leg-assignment and sign branches above. Put

$$\text{ht}[p : q] = \max(|p|, |q|), \quad h(\Gamma) = \text{ht}[p : q] \text{ht}[u : v] r.$$

For a globally realizable reduced hypotenuse C , define

$$h(C) = \min_{\Gamma \mapsto C} h(\Gamma). \quad (49)$$

A397123 orders the distinct globally realizable reduced hypotenuses first by increasing $h(C)$ and then numerically.

Lemma 4.6 (Finiteness). *For every H , only finitely many certificates satisfy $h(\Gamma) \leq H$.*

Proof. If $h_1 = \text{ht}[p : q]$ and $h_2 = \text{ht}[u : v]$, then $h_1 h_2 r \leq H$ implies $h_1, h_2, r \leq H$. There are only finitely many primitive projective pairs of bounded height and only finitely many s with $0 < s < r$. The remaining branch choices are finite. $\square$

Theorem 4.7 (Exact bounded-height enumeration). *For every positive integer H , the routine $\text{AList}(H)$ returns exactly the distinct globally realizable reduced hypotenuses C with $h(C) \leq H$, ordered by $(h(C), C)$.*

Proof. By Lemma 4.6, only finitely many primitive tuples (p, q, u, v, r, s) can have certificate height at most H , and the enumeration visits all of them. Conversely, every admissible realization determines two rational edge directions with square Euclidean norm and, by (36), square Minkowski norm. Lemma 4.1 puts each in projective normal form. After normalizing the first direction, the relative rational z -rotation of the second is one of (39)–(40). The primitive distance triangle has the unique Euclid parametrization (41) up to leg interchange. Hence the certificate search covers every admissible realization.

For each tuple, both rotation branches, both target-leg assignments, and the required sign branches are checked. The tests (38) and (43) are exact necessary rejection tests only. Every survivor is reconstructed by solving (48) exactly, every rational root is subjected to the Gaussian-gcd, parity and canonicity tests, and every accepted branch is finally verified from the three original points and their integer distances. Thus every returned value has a valid certificate of height at most H .

Conversely, if $h(C) \leq H$, choose a valid certificate of minimal height. Its primitive tuple and finite branch choices occur in the enumeration, so its exact reconstruction is reached and C is recorded. Keeping the smallest recorded height for each value and sorting $(h(C), C)$ gives precisely the stated order. $\square$

The current certified computation is complete through $H = 20000$ and yields 25 values. The permanent archive contains the exact Maple implementation and an independent exact C++ implementation whose recorded output agrees term-for-term and height-for-height on this full range. Its beginning is

17, 41, 97, 521, 233, 493, 697, 3457, 2813, 2417, 3001, 6757,
29857, 20213, 39761, 38497, 87029, 72689, 70093, 101393, . . .

Selected minimal certificates are

n	C	$h(C)$	$[p : q]$	$[u : v]$	(r, s)	reduced triangle
1	17	44	$[1 : 1]$	$[2 : 11]$	$(4, 1)$	$(8, 15, 17)$
2	41	55	$[1 : 1]$	$[2 : 11]$	$(5, 4)$	$(9, 40, 41)$
3	97	180	$[1 : 4]$	$[4 : 5]$	$(9, 4)$	$(65, 72, 97)$
4	521	220	$[1 : 1]$	$[2 : 11]$	$(20, 11)$	$(279, 440, 521)$
5	233	260	$[1 : 4]$	$[4 : 5]$	$(13, 8)$	$(105, 208, 233)$
6	493	440	$[1 : 4]$	$[4 : 5]$	$(22, 3)$	$(132, 475, 493)$
25	881	19525	$[2 : 11]$	$[71 : 49]$	$(25, 16)$	$(369, 800, 881)$

The exact Maple implementation is supplied as `AList_H_maple_oeis_final.mpl`. The permanent computational archive also contains an independent exact C++ implementation and its recorded $H = 20000$ output. The archive is reserved at Zenodo DOI 10.5281/zenodo.22666219 and contains a SHA-256 manifest for all source and output files.

5 A broken trail: 493 lies outside the quartic

Proposition 5.1. *The value 493 belongs to $\mathcal{S}_{123}$ but not to $\mathcal{S}_{677}$.*

Proof. Consider the three primitive canonical Euclid pairs

$$(19519, 18182), \quad (14206, 2243), \quad (22831, 15698).$$

They have gcd 1, opposite parity in each pair, and $m > k > 0$. The three pairwise distances of the associated Pythagorean points are

$$832812750, \quad 231434280, \quad 864371970.$$

Their common gcd is 1753290, and division gives

$$(475, 132, 493).$$

Indeed $475^2 + 132^2 = 493^2$ and $\gcd(475, 132) = 1$. Hence $493 \in \mathcal{S}_{123}$.

On the other hand, $493 = 17 \cdot 29$. If $493 \in \mathcal{S}_{677}$, Proposition 3.2 would force the nonexceptional prime 29 to satisfy $29 \equiv 1 \pmod{8}$, but $29 \equiv 5 \pmod{8}$. Therefore $493 \notin \mathcal{S}_{677}$. $\square$

Consequently

$$\boxed{\mathcal{S}_{123} \not\subseteq \mathcal{S}_{677}}. \quad (50)$$

There is also a global realization of 521 outside the original special linear family: the Euclid pairs

$$(3269, 1860), \quad (3458, 1023), \quad (4171, 1860)$$

have distances reducing to $(279, 440, 521)$. Thus “not in A399309” and “not globally realizable” are genuinely different conditions.

6 The mod-8 shadow: what survives after the break

Every primitive Pythagorean hypotenuse has all prime factors $1 \pmod{4}$, hence is 1 or 5 $\pmod{8}$. Let c_n be the n -th term of A008846 and define

$$b(n) = \begin{cases} 0, & c_n \equiv 1 \pmod{8}, \\ 1, & c_n \equiv 5 \pmod{8}. \end{cases} \quad (51)$$

Equivalently,

$$b(n) = \left(\frac{c_n - 1}{4} \right) \bmod 2 = \frac{1 - \left(\frac{2}{c_n} \right)}{2}, \quad (52)$$

where the last symbol is the Jacobi, equivalently Kronecker, symbol here. In prime-factor language,

$$b(n) \equiv \sum_{p \equiv 5 \pmod{8}} \nu_p(c_n) \pmod{2}.$$

The three geometric layers meet these bits differently:

- every value of $\mathcal{S}_{309}$ has bit 0 by Theorem 2.7;
- the quartic layer has both bits, for example 17 and $901 = 17 \cdot 53$;
- the global layer has both bits, for example 17 and 493.

Thus the failure of a global $1 \pmod{8}$ law is structural, not an isolated numerical anomaly.

7 Two realizability questions and the stress test 901

The preceding sections leave two logically distinct universality questions. The first concerns the relation between the quartic arithmetic envelope and the full global geometry.

Conjecture 7.1 (Quartic-to-global realizability). *Every primitive Pythagorean hypotenuse arising from a rational point on the quartic (12) is globally realizable by three primitive canonical Pythagorean points with pairwise integral distances forming a right triangle. Equivalently,*

$$\boxed{\mathcal{S}_{677} \subseteq \mathcal{S}_{123}.} \quad (53)$$

If Conjecture 7.1 holds, then the first separation by 521 and the reverse separation by 493 give the strict chain

$$\boxed{\mathcal{S}_{309} \subsetneq \mathcal{S}_{677} \subsetneq \mathcal{S}_{123}.}$$

The conjecture is stronger than rational reconstructibility. A global realization must simultaneously produce three primitive spinors, each of opposite parity, and after common symmetries all three must lie in the canonical chamber. The certificate equations are explicit, but uniform control of the Gaussian gcd and parity/canonicity conditions is the main obstruction.

The first four quartic values

$$17, \quad 41, \quad 521, \quad 881$$

are globally realized, with minimal certificate heights

$$44, \quad 55, \quad 220, \quad 19525.$$

The next quartic value is 901. Theorem 4.7 and the exact execution through $H = 20000$ show that 901 has no global certificate of height at most 20000. Therefore Conjecture 7.1, if true, already forces

$$h(901) > 20000. \quad (54)$$

This statement is not evidence that 901 is non-realizable. Rather, it shows that numerical size in the elliptic enumeration and global certificate complexity can be very far apart.

The second universality question is broader and independent of the quartic.

Question 7.2 (Global versus classical universality). *Is every primitive Pythagorean hypotenuse globally realizable in the sense of Definition 1.1? Equivalently, is*

$$\boxed{\mathcal{S}_{123} = \mathcal{H}?$$

At present neither equality nor strict containment is proved. This distinction is important when interpreting A397123. If the containment $\mathcal{S}_{123} \subseteq \mathcal{H}$ is strict, A397123 detects a genuinely new geometric subset of primitive Pythagorean hypotenuses. If equality holds, A397123 remains a different sequence because it orders those hypotenuses by the minimal global certificate height $h(C)$ rather than by numerical size. Thus its unconditional content is a realization-complexity invariant, while the possible strictness of the value set is an additional open problem.

8 What is proved, what is computed, and how to reproduce it

The logical status of the main ingredients is as follows.

Claim	Status	Verification in this paper / ancillary file
Special orthogonal family and quartic	exact algebra	Sections 2.1–2.2
Quartic–elliptic birational map	exact algebra	explicit maps in Section 2.3
Rank 3, torsion and full MW basis	computer-assisted exact theorem	Section 2.4 and the permanent certificate archive
A399309 finite search	theorem with exact computer-assisted input	Theorems 2.3 and 2.5, plus the permanent certificate archive
A399309 mod-8 law	local theorem using certified MW basis	Section 2.6
A399677 finite search	theorem with exact computer-assisted input	Theorem 2.5 and the permanent certificate archive
A397123 bounded-height completeness	algebraic theorem; finite data computer-assisted	Section 4; exact Maple source and independent C++ reproduction
$\mathcal{S}_{123} \subseteq \mathcal{H}$	exact from definitions	Definition 1.1
$493 \in \mathcal{S}_{123} \setminus \mathcal{S}_{677}$	exact hand calculation	Proposition 5.1
Quartic-to-global inclusion	open	Conjecture 7.1
Global equality $\mathcal{S}_{123} = \mathcal{H}$	open	Question 7.2

The currently certified finite data are logically separate from the conjecture:

- A399309 has 1442 distinct terms not exceeding 10^{400} in the current exact data set;
- A399677 is exhaustively enumerable below any prescribed numerical bound by the same Mordell–Weil theorem after removing the special-family filter;
- A397123 is currently exhaustively enumerated through certificate height 20000, yielding 25 values;
- A399705 is elementary once A008846 is known; its first 10000 bits have been tabulated.

None of these finite data sets proves Conjecture 7.1 or answers Question 7.2.

9 Where the search now stands

The same primitive Pythagorean parametrization appears twice. First it produces integral points on a quadratic cone. Then, after three such points are joined and their distances are

primitively reduced, it produces a second Pythagorean triangle. The interaction between these two layers generates a genus-one quartic, a rank-3 elliptic curve, local congruence restrictions and a global certificate-height theory.

The established relations are

$$\mathcal{S}_{309} \subsetneq \mathcal{S}_{677}, \quad \mathcal{S}_{309} \subsetneq \mathcal{S}_{123}, \quad \mathcal{S}_{123} \not\subseteq \mathcal{S}_{677}, \quad \mathcal{S}_{123} \subseteq \mathcal{H}.$$

Two questions remain open:

$$\mathcal{S}_{677} \stackrel{?}{\subseteq} \mathcal{S}_{123}, \quad \mathcal{S}_{123} \stackrel{?}{=} \mathcal{H}.$$

Thus the special family is a proper subset of both larger constructions; 493 proves that the global geometry is not contained in the quartic envelope; quartic-to-global realizability remains open; and it is also unknown whether the global value set is a proper subset of all primitive Pythagorean hypotenuses.

The arithmetic core of the paper is therefore independent of either open universality question. A399309 records the outputs of a concrete linear three-point construction; A399677 records the primitive hypotenuses on the associated genus-one quartic and inherits a rank-3 Mordell–Weil description and local prime restrictions; A397123 records the unrestricted global geometry through the minimal certificate height $h(C)$. Whether A397123 also defines a proper new value set inside A008846 is deliberately left open.

The value 901 remains a useful stress test for the first universality question, not a prerequisite for the validity of the preceding results. The paper is intended to separate these exact arithmetic and geometric structures from the genuinely harder rational-point questions that remain beyond them.

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